Power

M2
Module-power_12and5V-0.2
+12V_FROM_MAIN_RELAY
IN_CONSTANT_POWER
IN_VIGN
PWR_EN
+12V
VBAT
VIGN
+5V
+5VP
PG_5VP
GND

Bus types
General purpose bus
Ignition
Injection
Analog inputs (sensors)
Discrete inputs
LS & HS outputs
DC Motor

M4
Module-motor-driver-0.1
DC driver
OUT_ETB1
OUT_ETB2
+12V_FROM_MAIN_RELAY
+3V3
ETB_DIS
ETB_PWM
ETB_DIR
SPI3_CS_ETB

M7
Module-motor-driver-0.1
OUT_WASTEGATE1
OUT_WASTEGATE2
WASTEGATE_DIS
WASTEGATE_PWM
WASTEGATE_DIR
SPI3_CS_WASTEGATE

M3
Module-mc33810-0.1
OUT_IGN1
OUT_IGN2
OUT_IGN3
OUT_IGN4
OUT_INJ1
OUT_INJ2
OUT_INJ3
OUT_INJ4
+12V_FROM_MAIN_RELAY
+3.3VA
IGN1
IGN2
IGN3
IGN4
SPI3_MOSI
SPI3_SCK
SPI3_CS_33810
SPI3_MISO
-OUTEN
-SPKOUR
NOMI
MAXI
GND

LS4 FLYBACK1
Sidehold capable
OUT_CVVT1
OUT_CVVT2
OUT_WGT
OUT_RCV
OUT_AC_RELAY
OUT_AUX_LS1
OUT_AUX_LS2
OUT_CPSV_EVAP_PURGE
OUT_HEATER_CONTROL
12V_FLYBACK
CVVT1
CVVT2
WGT
RCV
AC_RELAY
AUX_LS1
AUX_LS2
CPSV_EVAP_PURGE
HEATER_CONTROL

LS6
Always hot load capable
OUT_CHECK_ENGINE
OUT_TACH
OUT_MAIN_RELAY
OUT_FAN_RELAY_LOW
OUT_FAN_RELAY_HIGH
OUT_FUEL_PUMP_RELAY
CHECK_ENGINE
TACH
MAIN_RELAY
FAN_RELAY_LOW
FAN_RELAY_HIGH
FUEL_PUMP_RELAY

M1
Module-mega-mcu144-0.6
x50 cant and accelerometer on SPI1 bus
Barometer on this I2C bus
FAQs does not have UART0
Internal TC2030 connector
Internal buttons
SPI2_SCK/CAN2_TX
SPI2_MISO
SPI2_MOSI
SPI2_CS/CAN2_RX
CPSV_EVAP_PURGE
HEATER_CONTROL
SPI3_SCK
SPI3_MISO
SPI3_MOSI
SPI3_CS
CANH
CANL
I2C_SDA
I2C_SCL
UART0_RX
UART0_TX
UART2_RX
UART2_TX
USBP
USBM
USBID
VBUS
SWCLK
SWDIO
SWO
nReset
BOOT0
SPI3_CS_ETB
SPI3_CS_WASTEGATE
PG_5VP
-SPKOUR
MAXI
NOMI
-OUTEN
SPI3_CS_33810
V33_SWITCHABLE
+3.3VA
+3V3
+5VA
VSA_SWITCHABLE
GNDA
IN_CRANK
IN_CAM
IN_VSS
IN_TPS
IN_TPS2
IN_PPS1
IN_PPS2
IN_IAT
IN_TEMP
IN_CLT
IN_OILT
IN_AC_ON
IN_CLUTCH
IN_BRAKE
IN_MAP1
IN_MAP2
IN_AC_PRESSURE
IN_WBO
IN_BK1_O2_NARROW
IN_WASTEGATE_POS
IN_PS_SWITCH
DIN1
DIN2
DIN3
KNOCK
IN_AUX1
IN_AUX2
IN_AUX3
IN_AUX4
IN_SENS1
IN_SENS2
IN_SENS3
IN_SENS4
IN_KNOCK
IN_D1
IN_D2
IN_D3
IN_D4
CAN_WUPMD
VREF1
VREF2
V33_REF
OUT_PWR_EN
IN_VIGN
VBAT
VCC
GND

M6
M5
Module-can-0.1
Module-wbo-0.5
CAN2_RX
CAN2_TX
CAN_VIO
CAN_RX
CAN_TX
CANL
CANH
+5V
+3V3
GND
CAN_VIO
CANL
CANH
V33_OUT
SWCLK
SWDIO
nReset
VDDA
VS_IN
Power GND
Heater negative
LSU_nernst
LSU_Rtrim
LSU_ip
LSU_vm
LSU_un
Analog/digital GND
PULL_UP1
SEL1
PULL_DOWN1
PULL_UP2
SEL2
PULL_DOWN2
+5VA
GND
WBO_Heater
WBO_Rtrim
WBO_ip
WBO_Vm
WBO_un
+12V_FROM_MAIN_RELAY
39281063

USB_B
J40
VBUS
USB+
USB-
GND
Shield
J1
Conn_01x05_Male

P1B
284617-1
IN_VIGN
GND
+12V_FROM_MAIN_RELAY
IN_CONSTANT_POWER
IN_MAP2
IN_KNOCK_RAW
IN_CAM2
IN_CRANK
IN_TEMP
IN_WASTEGATE_POS
OUT_INJ1
OUT_INJ2
OUT_INJ3
OUT_INJ4
IN_MAP1
IN_TPS2
IN_CLT
IN_PPS2
IN_BK1_O2_NARROW
IN_VSS
P02
+5VP
Alternator
OEM_BK1
OUT_WGT
OUT_RCV
P10 VIS
IN_IAT
IN_AC_PRESSURE
P11 Blower motor
IN_OILT
IN_CAM1
OUT_MAIN_RELAY
OUT_FAN_RELAY_LOW
OUT_CVVT1
OUT_CPSV_EVAP_PURGE
OUT_CVVT2
P13 IMMO
OUT_FUEL_PUMP_RELAY
OUT_ETB2
OUT_ETB1
P14
P15 IMMO
P16 DATA
CAN+
CAN-
IN_TPS1
IN_PPS1
IN_WBO
used by Kia forte
OUT_TACH
OUT_AC_RELAY
OUT_FAN_RELAY_HIGH
P19 CRUISE
P20
OUT_CHECK_ENGINE
OUT_HEATER_CONTROL
P22 O2

P1A
284617-1
OUT_IGN4
P23
P24
P25
P26
P27
P28
P29
P30 Wiper SW
P31 Generator FR
OUT_IGN2
P32
P33
P34
P35
P36
P37
P38
IN_BRAKE
P39 cruise
OUT_IGN1
P40
P41
P42
P43
P44
P45
P46
P47
P48
IN_CLUTCH
P49 Brake test
OUT_WASTEGATE2
OUT_IGN3
P50
P51
P52
P53
P54
P55
IN_AC_ON first generation BK1
IN_PS_SWITCH
OUT_WASTEGATE1

J2
39281123
OUT_AUX_LS3
OUT_AUX_LS2
DIN3
AIN3
OUT_AUX_LS1
DIN1
AIN2
AIN1
+12V_FROM_MAIN_RELAY
+5VP
R1 R2 R3 R4 R5 R6 GNP
AIN1 R11 R12 R13 R14 R15 R16 GNDA
AIN2 R21 R22 R23 R24 R25 R26 GNDA
DIN1 R31 R32 R33 R34 R35 R36 GNDA
DIN2 R41 R42 R43 R44 R45 R46 GNDA
DIN3 R51 R52 R53 R54 R55 R56 GNDA

M9
Module-knock-0.2
+5VA
VREF2
VSA
VREF
GND
IN_KNOCK_RAW
IN_KNOCK
OUT_KNOCK

BT1
CR1220
D24
1N4148WS
+5V
GND

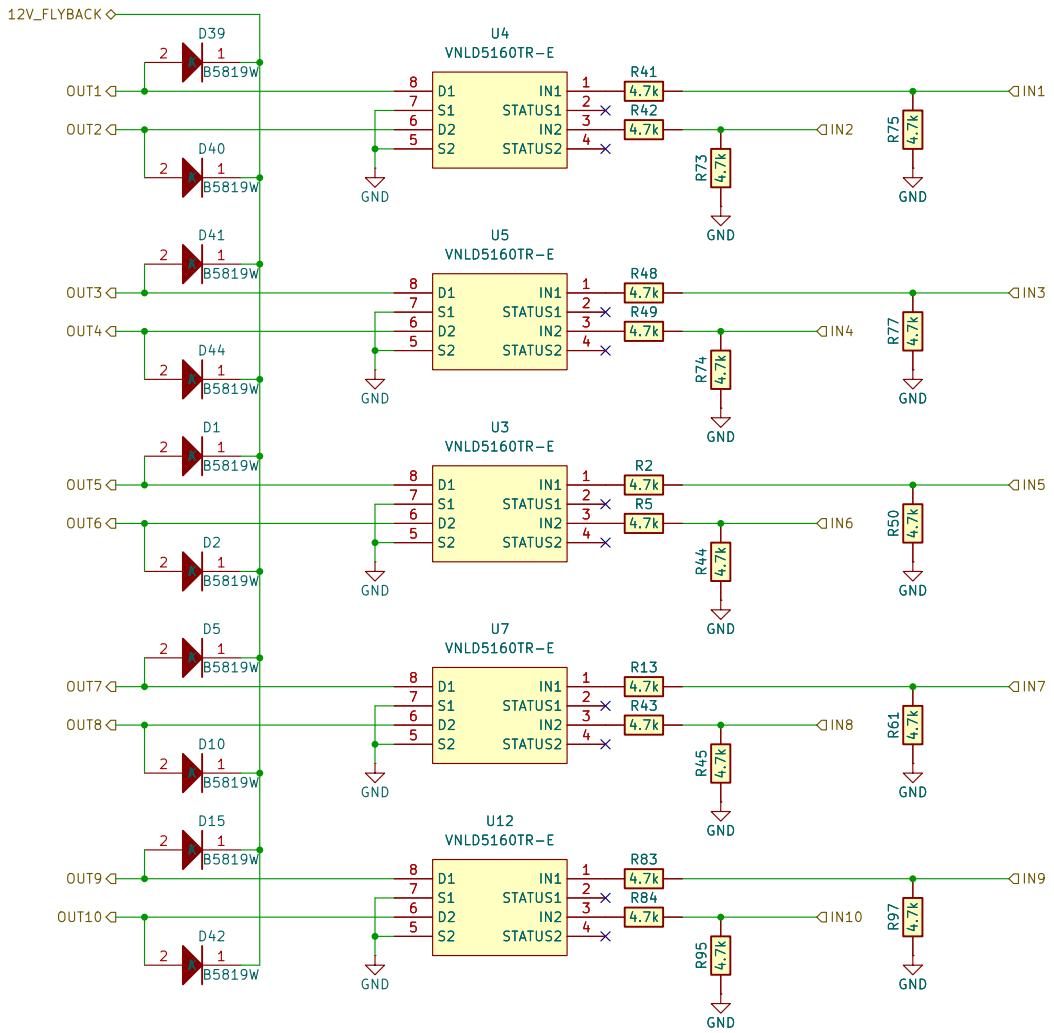
LEDs
R6 10k
D6 LED-RED-0603 IN4
D7 LED-RED-0603 IN3
D8 LED-RED-0603 IN2
D9 LED-RED-0603 IN1
+12V_FROM_MAIN_RELAY
R7 20k
D16 LED-WHITE-0603 1N4148WS OUT_TACH
D20 LED-WHITE-0603 1N4148WS OUT_FAN_RELAY_LOW
D21 LED-WHITE-0603 1N4148WS OUT_MAIN_RELAY
D19 LED-WHITE-0603 1N4148WS OUT_AC_RELAY
D18 LED-WHITE-0603 1N4148WS
D22
R76 20k
D36 LED-WHITE-0603 1N4148WS OUT_FUEL_PUMP_RELAY
D33 LED-WHITE-0603 1N4148WS OUT_CHECK_ENGINE
D34 LED-WHITE-0603 1N4148WS OUT_FAN_RELAY_HIGH
D38 LED-WHITE-0603 1N4148WS
D35
+5V
R10 220
D28 LED-WHITE-0603 1N4148WS IN_CRANK
D29 LED-WHITE-0603 1N4148WS IN_CAM2
D30 LED-WHITE-0603 1N4148WS IN_CAM1
D27

Prototype areas
+3V3
+5V
GND
+12V_FROM_MAIN_RELAY
PROTO_AREA_2
J3 Conn_01x04
TX1
RX1
TX2
RX2
J22
J24

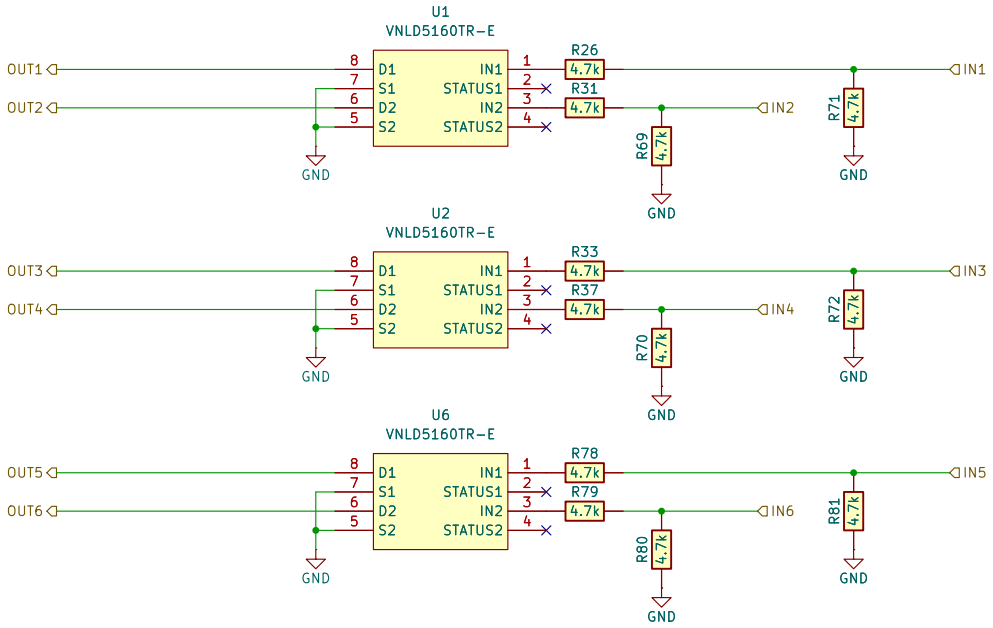
FOR OFF ROAD PURPOSES ONLY
This is not for applications with emissions or safety regulations (AKA not for street use). This is for closed stages, track and equipment.
https://rusefi.com/s/hellen154hyundai

Board Info
Hellen-One-PCB-logo

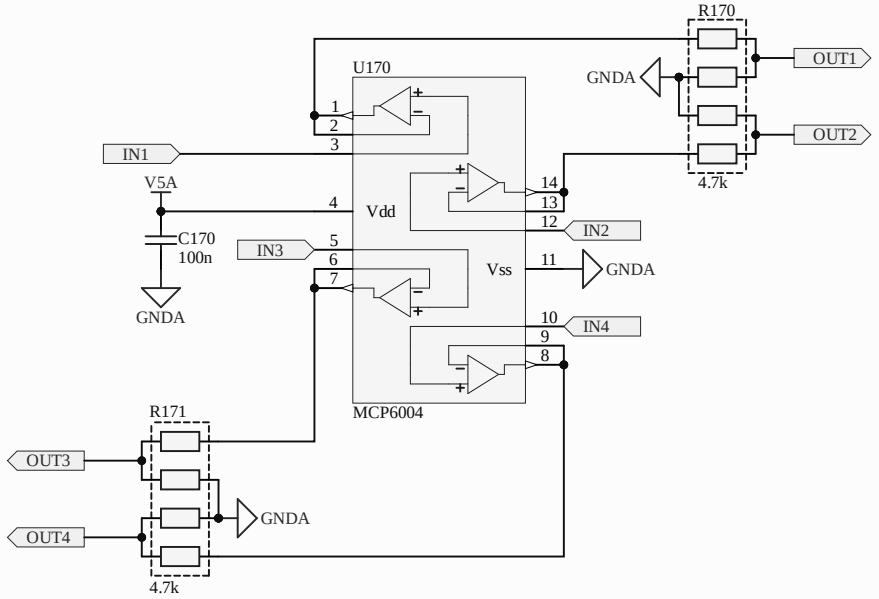
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Title: hellen154hyundai
Size: A1 | Date: 2024-05-13 | Rev: E
KiCad E.D.A. 10.0.8 | Id: 1/3

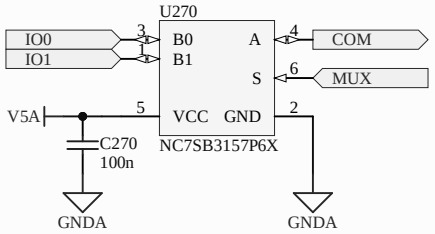


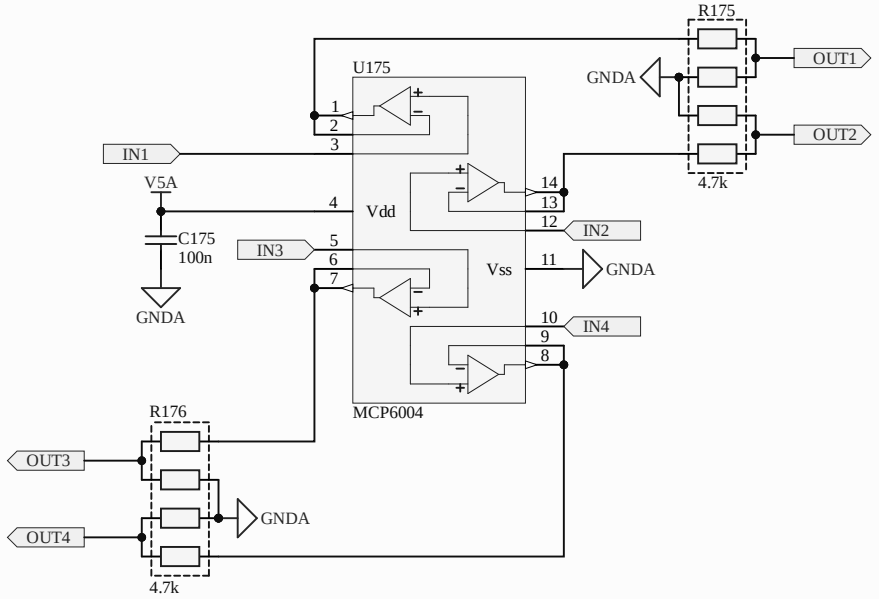
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Sheet: /LS4 FLYBACK1/		
File: LS4_FLYBACK.kicad_sch		
Title: hellen154hyundai		
Size: A4	Date: 2024-05-13	**Rev: E**
KiCad E.D.A. 10.0.8		Id: 2/3

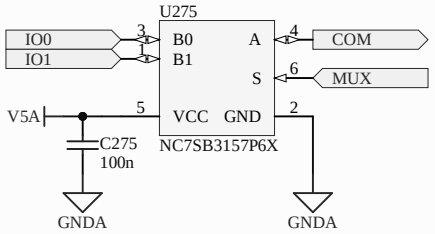


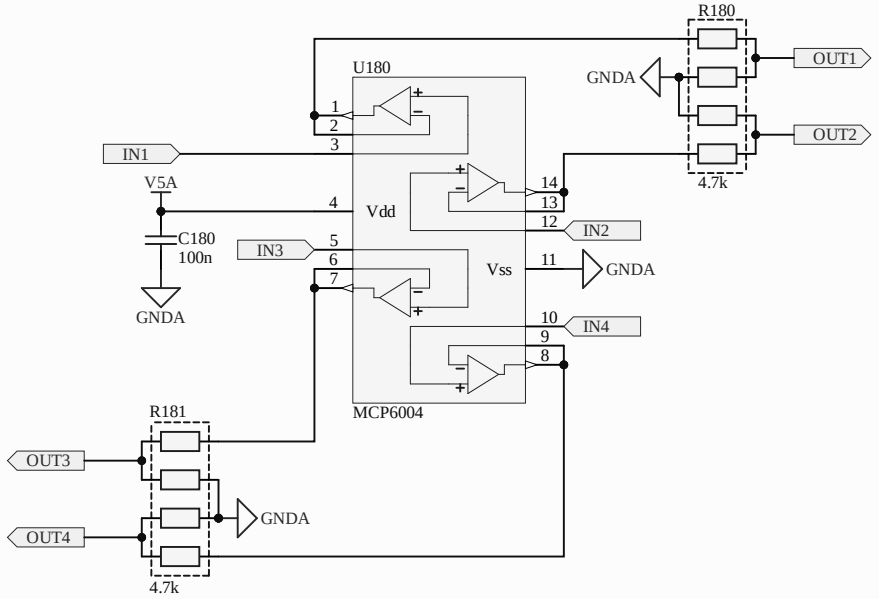
rusEFI		
Sheet: /LS6/		
File: LS6.kicad_sch		
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Size: A4	Date: 2024-05-13	**Rev: E**
KiCad E.D.A. 10.0.8		Id: 3/3

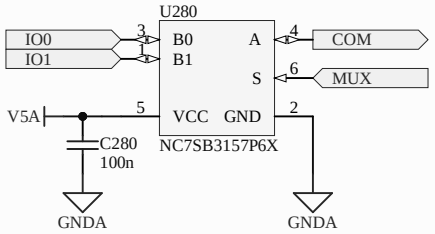

R170
U170
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C170
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R171
OUT3
GNDA
OUT4
4.7k

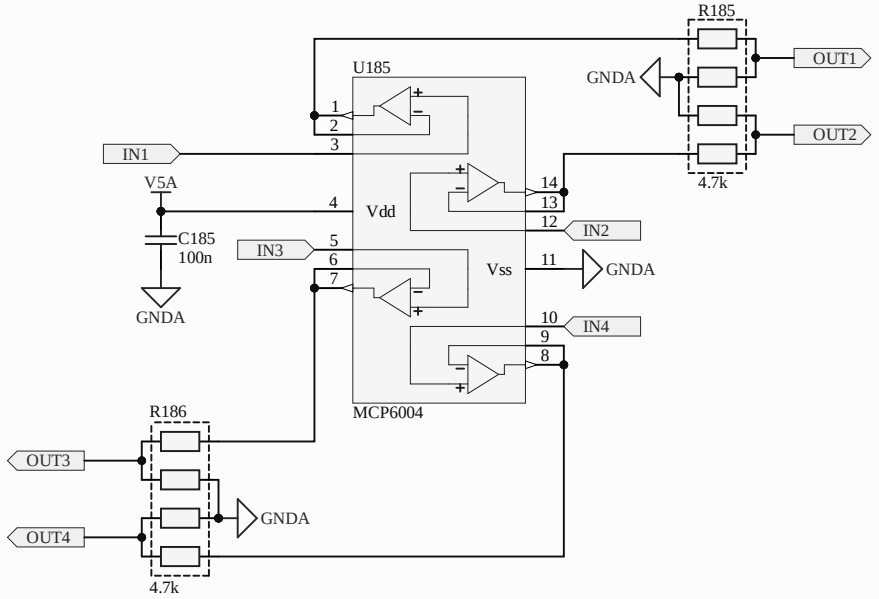

U270
IO0
IO1
3
1
B0
B1
A
S
4
COM
6
MUX
V5A
5
VCC
GND
2
C270
100n
NC7SB3157P6X
GNDA
GNDA

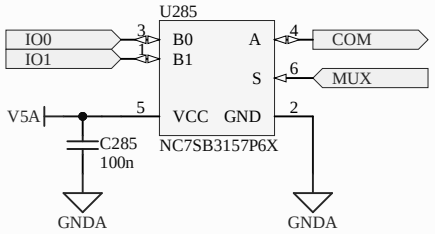

R175
U175
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C175
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R176
OUT3
GNDA
OUT4
4.7k

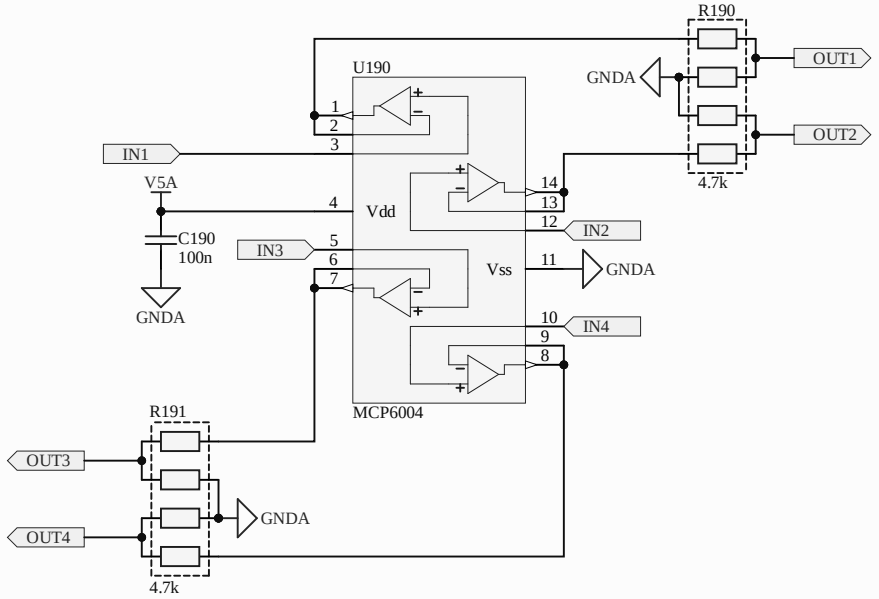

U275
IO0
IO1
3
1
B0
B1
A
S
4
6
COM
MUX
V5A
5
VCC
GND
2
C275
100n
NC7SB3157P6X
GNDA
GNDA

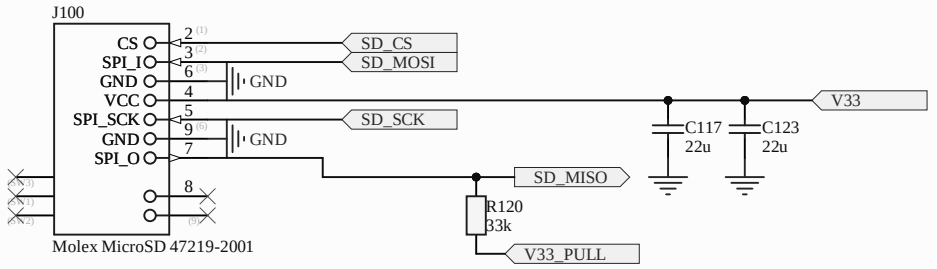

R180
U180
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C180
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R181
OUT3
GNDA
OUT4
4.7k

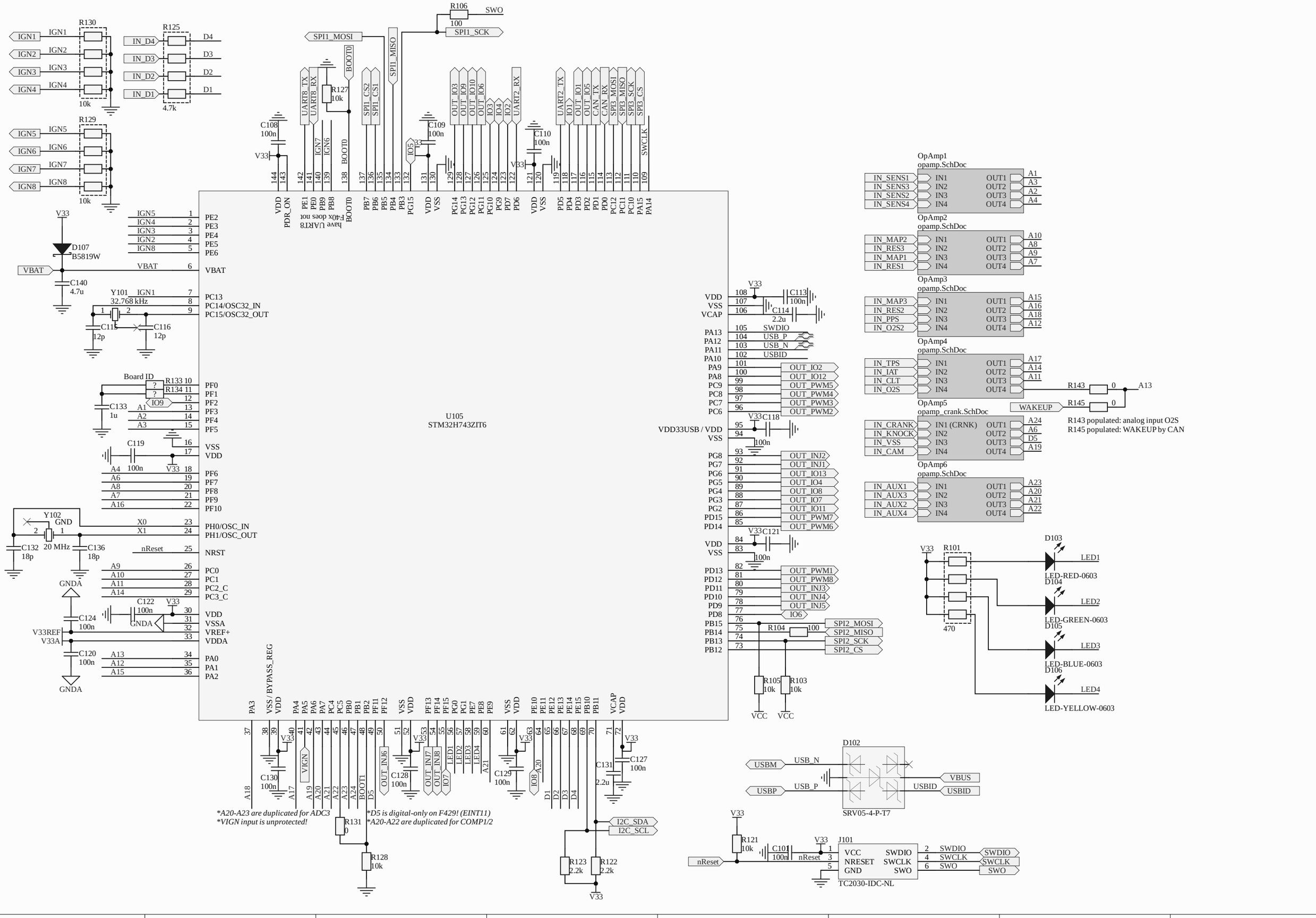

U280
IO0
IO1
3
1
B0
B1
A
S
4
COM
6
MUX
V5A
5
VCC
GND
2
C280
100n
NC7SB3157P6X
GNDA
GNDA

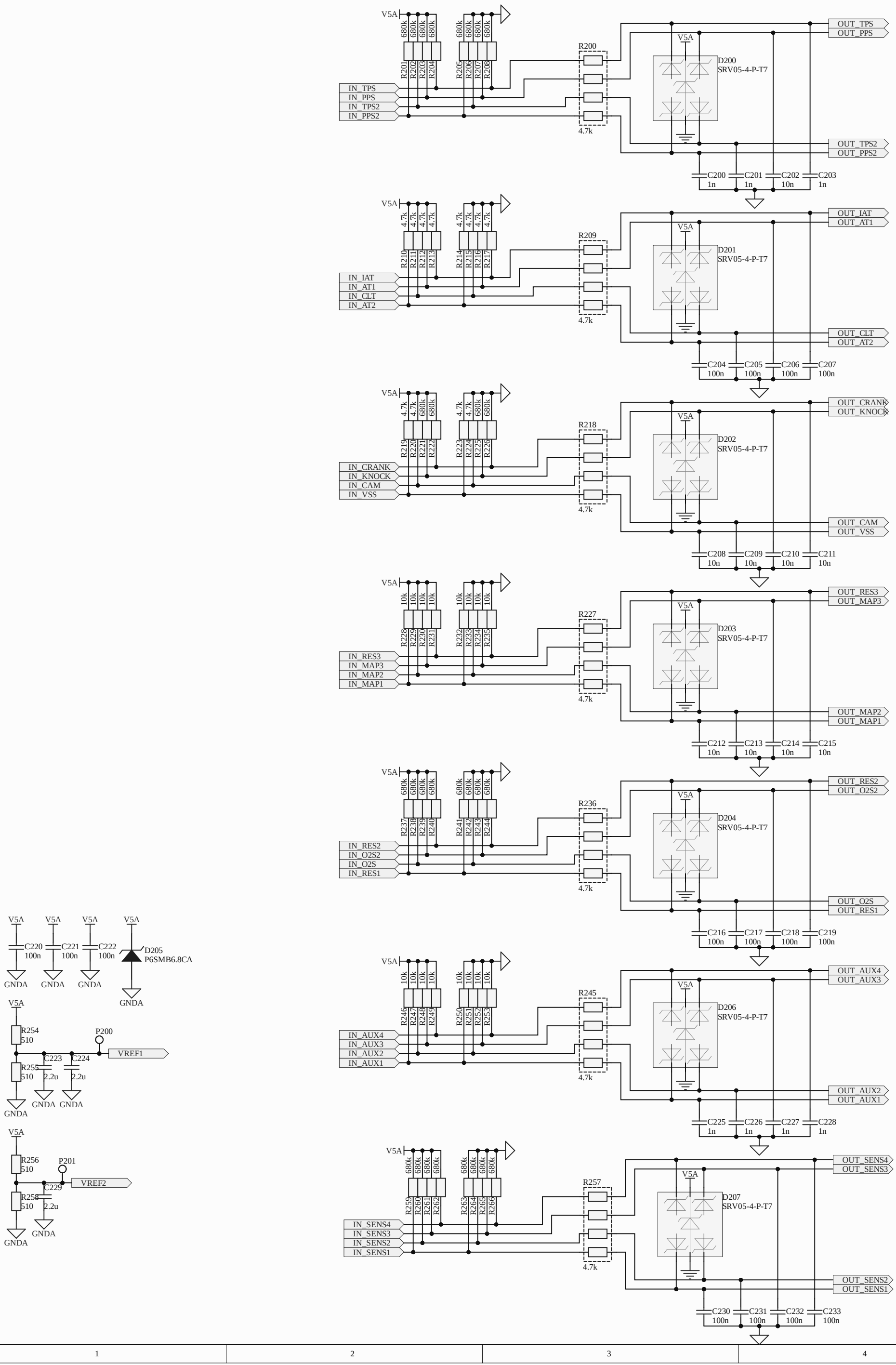

R185
U185
GNDA
OUT1
OUT2
IN1
V5A
4
Vdd
14
13
12
IN2
C185
100n
IN3
5
6
7
Vss
11
GNDA
GNDA
10
IN4
9
8
MCP6004
R186
OUT3
OUT4
GNDA
4.7k
4.7k

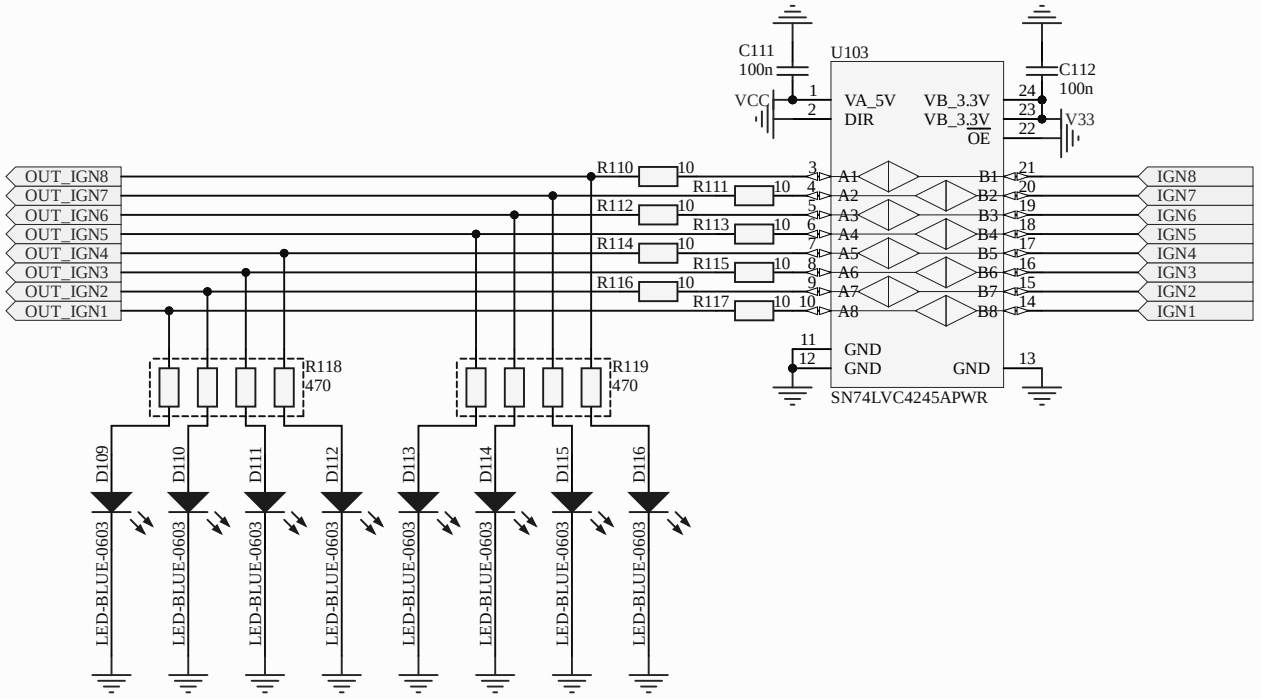

U285
IO0
IO1
3
1
B0
B1
A
S
4
COM
6
MUX
V5A
5
VCC
GND
2
C285
100n
NC7SB3157P6X
GNDA
GNDA

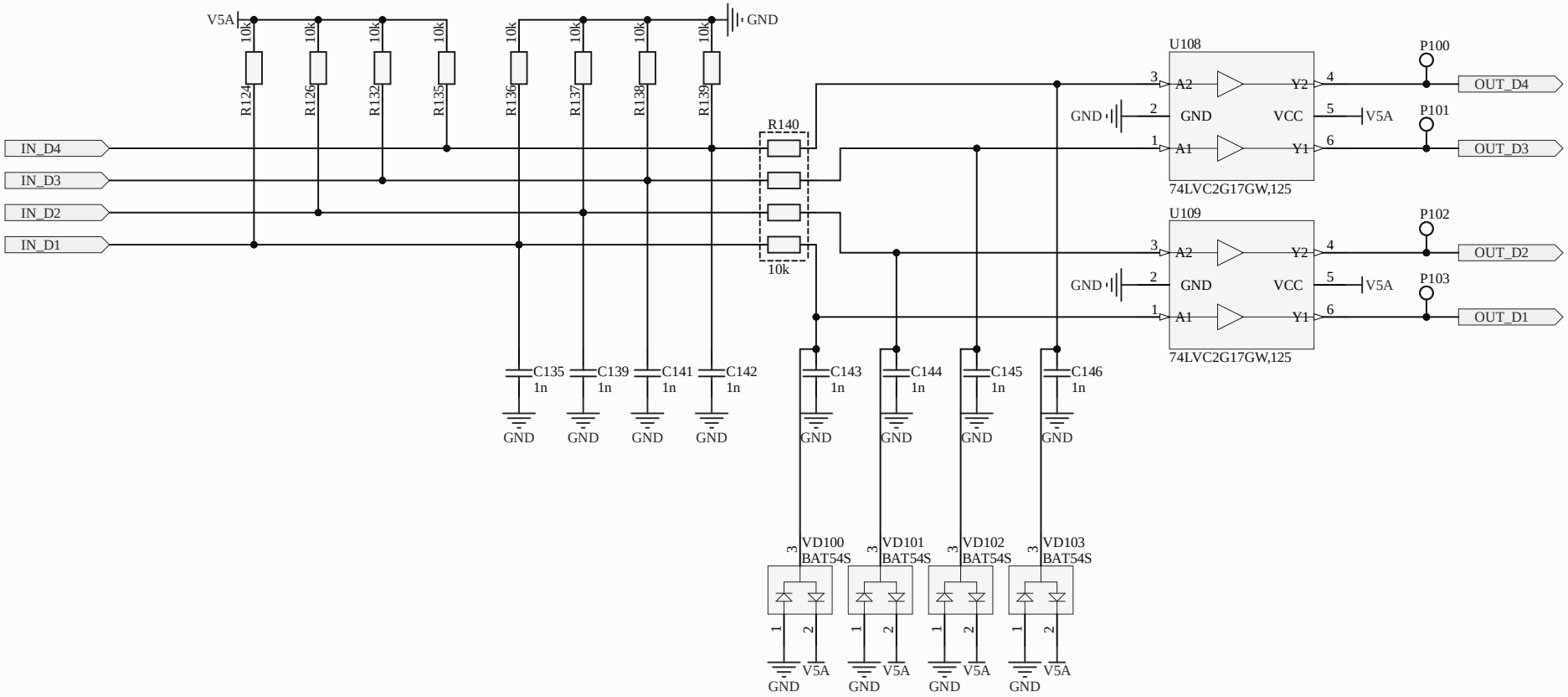

R190
U190
GNDA
OUT1
OUT2
IN1
V5A
Vdd
4.7k
C190
100n
IN3
IN2
Vss
GNDA
GNDA
IN4
MCP6004
R191
OUT3
GNDA
OUT4
4.7k

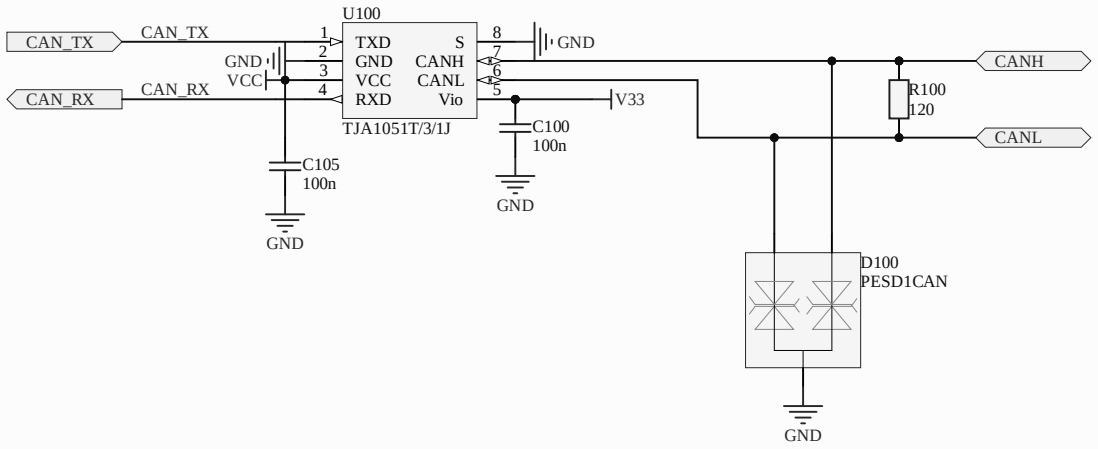

J100
CS
SPI_I
GND
VCC
SPI_SCK
GND
SPI_O
2
3
6
4
5
9
7
8
GND
GND
SD_CS
SD_MOSI
SD_SCK
SD_MISO
V33
C117
22u
C123
22u
R120
33k
V33_PULL
Molex MicroSD 47219-2001

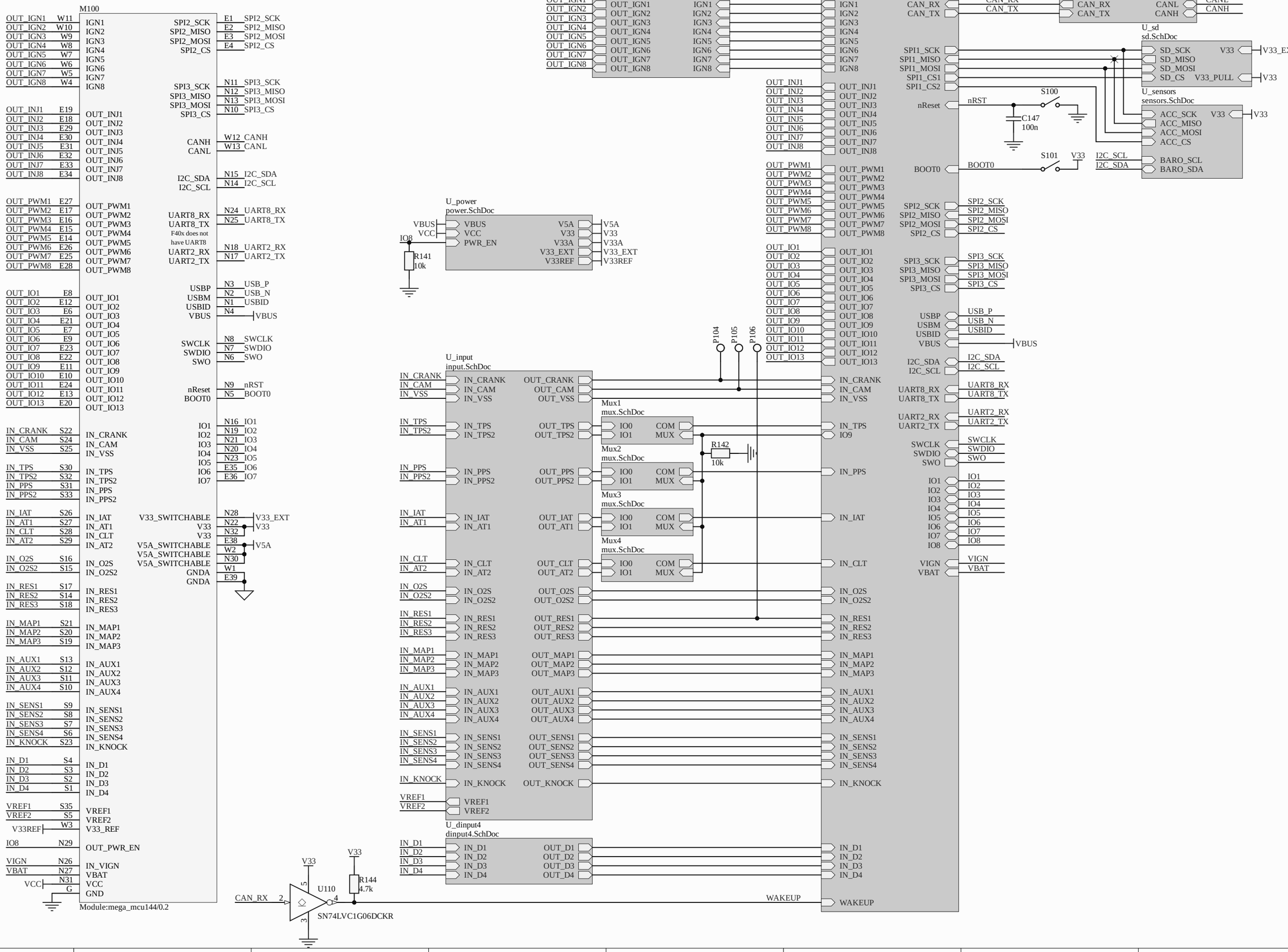

R130
R125
R129
IGN1
IGN2
IGN3
IGN4
IGN5
IGN6
IGN7
IGN8
IN_D4
IN_D3
IN_D2
IN_D1
D4
D3
D2
D1
10k
4.7k
R106
SWO
100
SPI1_SCK
SPI1_MOSI
SPI1_MISO
R127
10k
BOOT0
C108
100n
C109
100n
C110
100n
V33
UART8_TX
UART8_RX
SPI1_CS2
SPI1_CS1
OUT_IO3
OUT_IO9
OUT_IO10
OUT_IO6
IO3
IO4
IO2
UART2_RX
UART2_TX
IO1
OUT_IO1
OUT_IO5
CAN_TX
CAN_RX
SPI3_MOSI
SPI3_MISO
SPI3_SCK
SPI3_CS
SWCLK
VDD
PDR_ON
PE1
PE0
PB9
PB8
have UART8 F40x does not
BOOT0
PB7
PB6
PB5
PB4
PB3
PG15
PG14
PG13
PG12
PG11
PG10
PC9
PD7
PD6
PD5
PD4
PD3
PD2
PD1
PD0
PC12
PC11
PC10
PA15
PA14
VSS
D107
B5819W
VBAT
C140
4.7u
Y101
32.768 kHz
C115
12p
C116
12p
PE2
PE3
PE4
PE5
PE6
PC13
PC14/OSC32_IN
PC15/OSC32_OUT
Board ID
R133
R134
IO9
C133
1u
PF0
PF1
PF2
PF3
PF4
PF5
C119
100n
VSS
VDD
PF6
PF7
PF8
PF9
PF10
Y102
GND
20 MHz
C132
18p
C136
18p
PH0/OSC_IN
PH1/OSC_OUT
nReset
NRST
PC0
PC1
PC2_C
PC3_C
GNDA
C124
100n
C122
100n
VDDA
VSSA
VREF+
V33REF
V33A
C120
100n
PA0
PA1
PA2
U105
STM32H743ZIT6
VCAP
C113
100n
C114
2.2u
SWDIO
USB_P
USB_N
USBID
PA13
PA12
PA11
PA10
PA9
PA8
PC9
PC8
PC7
PC6
OUT_IO2
OUT_IO12
OUT_PWM5
OUT_PWM4
OUT_PWM3
OUT_PWM2
VDD33USB / VDD
C118
100n
PG8
PG7
PG6
PG5
PG4
PG3
PG2
PD15
PD14
OUT_INJ2
OUT_INJ1
OUT_IO13
OUT_IO4
OUT_IO8
OUT_IO7
OUT_IO11
OUT_PWM7
OUT_PWM6
C121
100n
PD13
PD12
PD11
PD10
PD9
PD8
PB15
PB14
PB13
PB12
OUT_PWM1
OUT_PWM8
OUT_INJ3
OUT_INJ4
OUT_INJ5
IO6
SPI2_MOSI
SPI2_MISO
SPI2_SCK
SPI2_CS
R104
100
R105
10k
R103
10k
VCC
OpAmp1
opamp.SchDoc
IN_SENS1
IN_SENS3
IN_SENS2
IN_SENS4
IN1
IN2
IN3
IN4
OUT1
OUT2
OUT3
OUT4
A1
A3
A2
A4
OpAmp2
IN_MAP2
IN_RES3
IN_MAP1
IN_RES1
A10
A8
A9
A7
OpAmp3
IN_MAP3
IN_RES2
IN_PPS
IN_O2S2
A15
A16
A18
A12
OpAmp4
IN_TPS
IN_IAT
IN_CLT
IN_O2S
A17
A14
A11
OpAmp5
opamp_crank.SchDoc
IN_CRANK
IN_KNOCK
IN_VSS
IN_CAM
IN1 (CRNK)
A24
A6
D5
A19
OpAmp6
IN_AUX1
IN_AUX3
IN_AUX2
IN_AUX4
A23
A20
A21
A22
R143
0
A13
WAKEUP
R145
R143 populated: analog input O2S
R145 populated: WAKEUP by CAN
D103
D104
D105
D106
LED1
LED2
LED3
LED4
LED-RED-0603
LED-GREEN-0603
LED-BLUE-0603
LED-YELLOW-0603
R101
470
D102
USBM
USBP
VBUS
SRV05-4-P-T7
R121
10k
C101
100n
J101
VCC
NRESET
GND
SWDIO
SWCLK
SWO
TC2030-IDC-NL
VIGN
OUT_INJ6
OUT_INJ7
OUT_INJ8
IO7
IO8
LED1
LED2
LED3
LED4
I2C_SDA
I2C_SCL
R131
0
R128
10k
R123
2.2k
R122
2.2k
C130
100n
C128
100n
C129
100n
C131
2.2u
C127
100n
*A20-A23 are duplicated for ADC3
*VIGN input is unprotected!
*D5 is digital-only on F429! (EINT11)
*A20-A22 are duplicated for COMP1/2

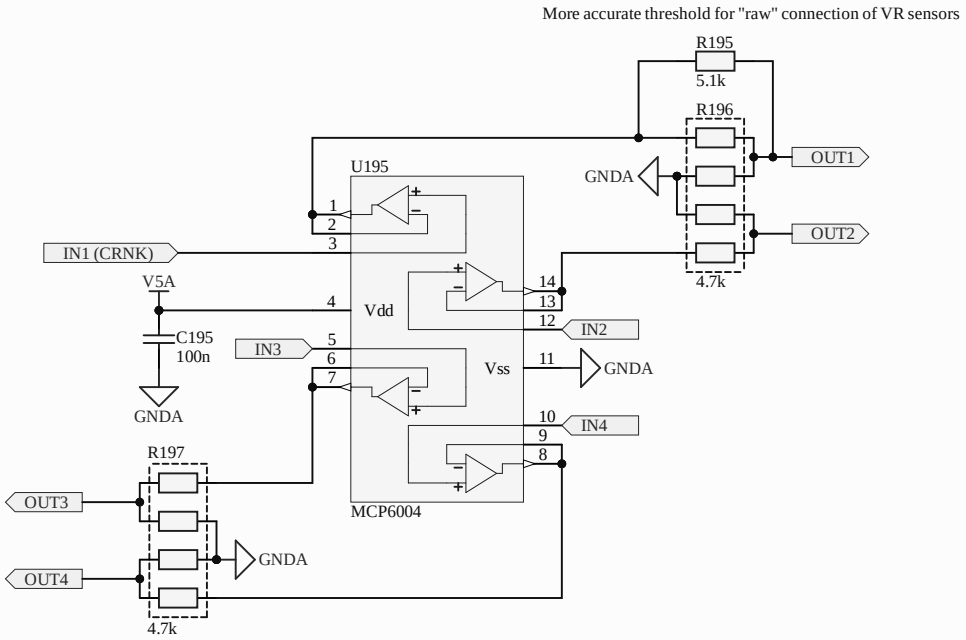

V5A
680k
R201
R202
R203
R204
R205
R206
R207
R208
IN_TPS
IN_PPS
IN_TPS2
IN_PPS2
R200
4.7k
D200
SRV05-4-P-T7
OUT_TPS
OUT_PPS
OUT_TPS2
OUT_PPS2
C200 1n
C201 1n
C202 10n
C203 1n
4.7k
R210
R211
R212
R213
R214
R215
R216
R217
IN_IAT
IN_AT1
IN_CLT
IN_AT2
R209
D201
OUT_IAT
OUT_AT1
OUT_CLT
OUT_AT2
C204 100n
C205 100n
C206 100n
C207 100n
R219
R220
R221
R222
R223
R224
R225
R226
IN_CRANK
IN_KNOCK
IN_CAM
IN_VSS
R218
D202
OUT_CRANK
OUT_KNOCK
OUT_CAM
OUT_VSS
C208 10n
C209 10n
C210 10n
C211 10n
10k
R228
R229
R230
R231
R232
R233
R234
R235
IN_RES3
IN_MAP3
IN_MAP2
IN_MAP1
R227
D203
OUT_RES3
OUT_MAP3
OUT_MAP2
OUT_MAP1
C212 10n
C213 10n
C214 10n
C215 10n
R237
R238
R239
R240
R241
R242
R243
R244
IN_RES2
IN_O2S2
IN_O2S
IN_RES1
R236
D204
OUT_RES2
OUT_O2S2
OUT_O2S
OUT_RES1
C216 100n
C217 100n
C218 100n
C219 100n
R246
R247
R248
R249
R250
R251
R252
R253
IN_AUX4
IN_AUX3
IN_AUX2
IN_AUX1
R245
D206
OUT_AUX4
OUT_AUX3
OUT_AUX2
OUT_AUX1
C225 1n
C226 1n
C227 1n
C228 1n
R259
R260
R261
R262
R263
R264
R265
R266
IN_SENS4
IN_SENS3
IN_SENS2
IN_SENS1
R257
D207
OUT_SENS4
OUT_SENS3
OUT_SENS2
OUT_SENS1
C230 100n
C231 100n
C232 100n
C233 100n
C220 100n
C221 100n
C222 100n
D205
P6SMB6.8CA
GNDA
R254 510
P200
VREF1
C223
C224
R255 510
2.2u
R256 510
P201
VREF2
C229
R258 510

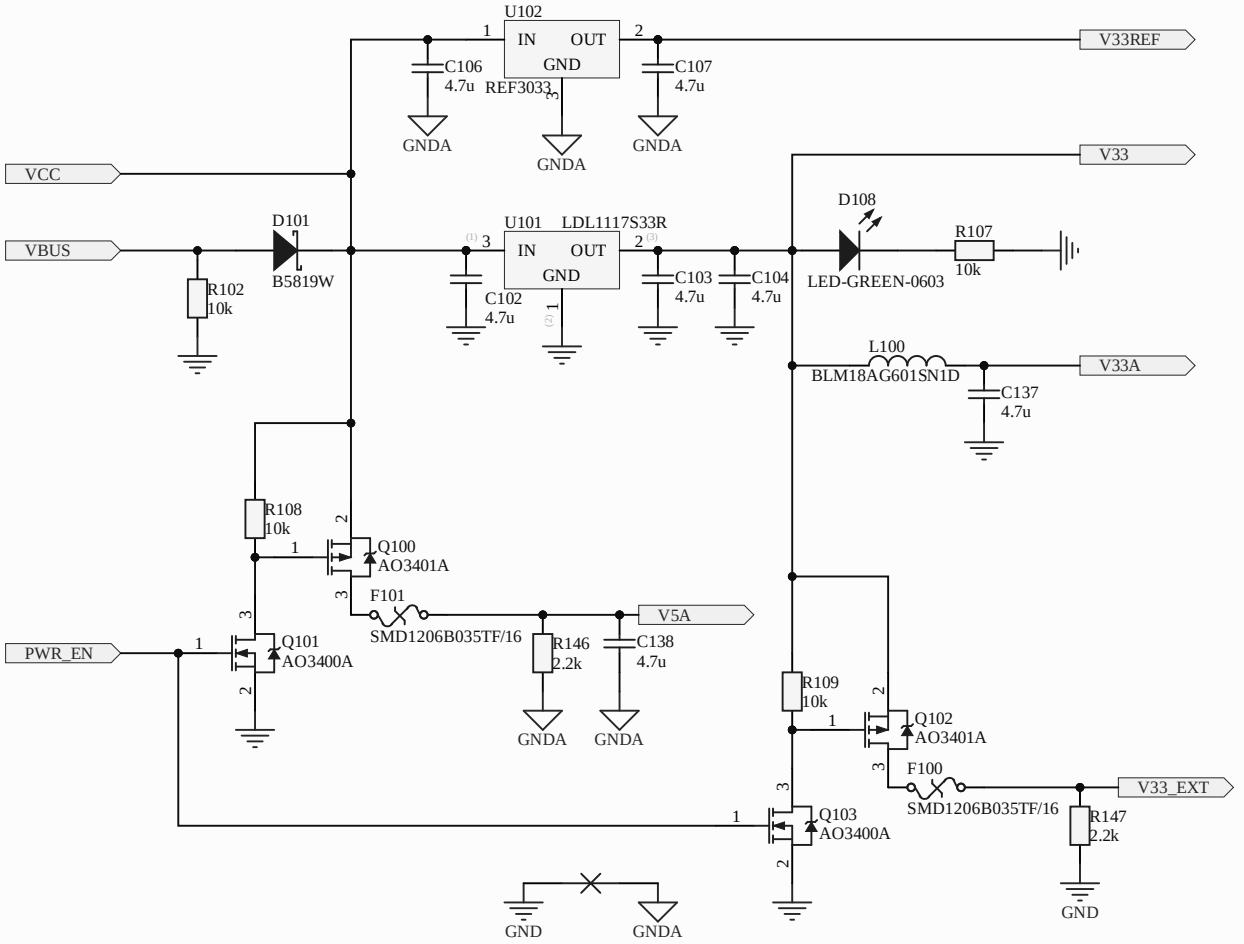

C111
100n
VCC
U103
VA_5V
DIR
VB_3.3V
VB_3.3V
OE
C112
100n
V33
GND
GND
GND
SN74LVC4245APWR
OUT_IGN8
OUT_IGN7
OUT_IGN6
OUT_IGN5
OUT_IGN4
OUT_IGN3
OUT_IGN2
OUT_IGN1
R110
R111
R112
R113
R114
R115
R116
R117
R118
470
R119
470
A1
A2
A3
A4
A5
A6
A7
A8
B1
B2
B3
B4
B5
B6
B7
B8
IGN8
IGN7
IGN6
IGN5
IGN4
IGN3
IGN2
IGN1
D109
D110
D111
D112
D113
D114
D115
D116
LED-BLUE-0603


V5A
GND
10k
R124
R126
R132
R135
R136
R137
R138
R139
R140
IN_D4
IN_D3
IN_D2
IN_D1
C135 1n
C139 1n
C141 1n
C142 1n
C143 1n
C144 1n
C145 1n
C146 1n
U108
U109
A2
A1
GND
VCC
Y2
Y1
74LVC2G17GW,125
P100
P101
P102
P103
OUT_D4
OUT_D3
OUT_D2
OUT_D1
VD100 BAT54S
VD101 BAT54S
VD102 BAT54S
VD103 BAT54S

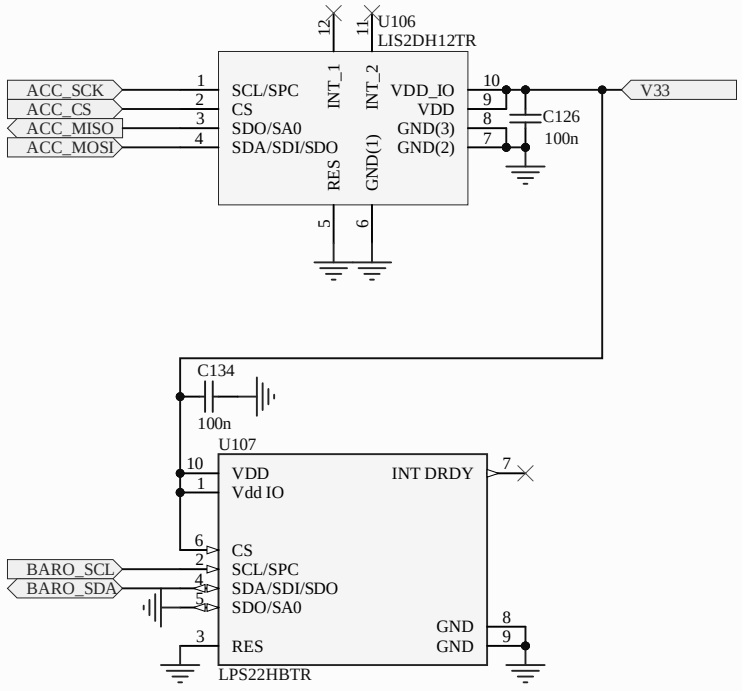

U100
CAN_TX
CAN_TX
CAN_RX
CAN_RX
GND
VCC
1
2
3
4
TXD
GND
VCC
RXD
S
CANH
CANL
Vio
8
7
6
5
GND
V33
CANH
CANL
R100
120
C105
100n
C100
100n
TJA1051T/3/1J
D100
PESD1CAN
GND
GND
GND

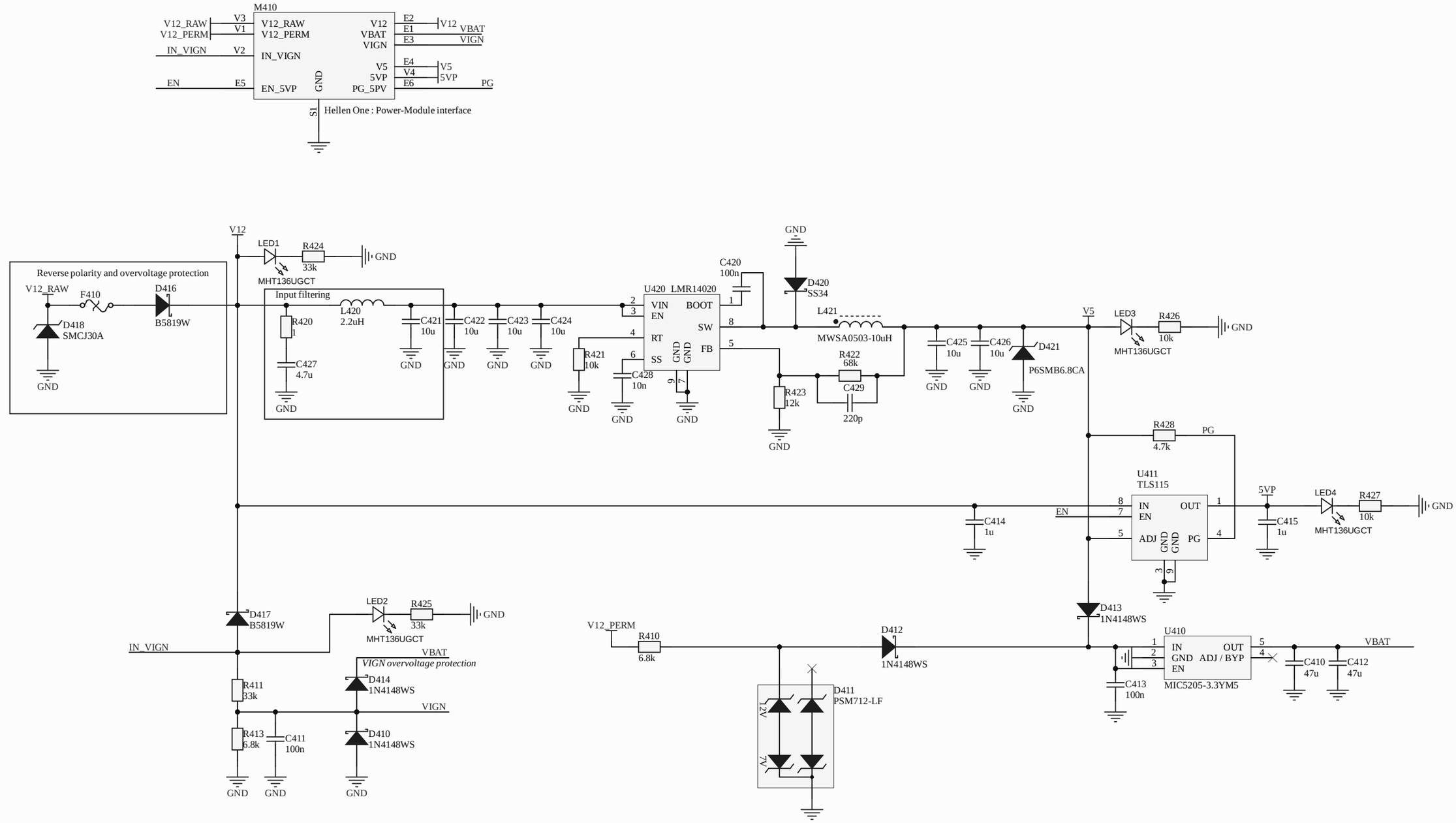

M100
OUT_IGN1 W11
OUT_IGN2 W10
OUT_IGN3 W9
OUT_IGN4 W8
OUT_IGN5 W7
OUT_IGN6 W6
OUT_IGN7 W5
OUT_IGN8 W4
IGN1
IGN2
IGN3
IGN4
IGN5
IGN6
IGN7
IGN8
SPI2_SCK E1 SPI2_SCK
SPI2_MISO E2 SPI2_MISO
SPI2_MOSI E3 SPI2_MOSI
SPI2_CS E4 SPI2_CS
SPI3_SCK N11 SPI3_SCK
SPI3_MISO N12 SPI3_MISO
SPI3_MOSI N13 SPI3_MOSI
SPI3_CS N10 SPI3_CS
OUT_INJ1 E19
OUT_INJ2 E18
OUT_INJ3 E29
OUT_INJ4 E30
OUT_INJ5 E31
OUT_INJ6 E32
OUT_INJ7 E33
OUT_INJ8 E34
OUT_INJ1
OUT_INJ2
OUT_INJ3
OUT_INJ4
OUT_INJ5
OUT_INJ6
OUT_INJ7
OUT_INJ8
CANH W12 CANH
CANL W13 CANL
I2C_SDA N15 I2C_SDA
I2C_SCL N14 I2C_SCL
OUT_PWM1 E27
OUT_PWM2 E17
OUT_PWM3 E16
OUT_PWM4 E15
OUT_PWM5 E14
OUT_PWM6 E26
OUT_PWM7 E25
OUT_PWM8 E28
OUT_PWM1
OUT_PWM2
OUT_PWM3
OUT_PWM4
OUT_PWM5
OUT_PWM6
OUT_PWM7
OUT_PWM8
UART8_RX N24 UART8_RX
UART8_TX N25 UART8_TX
F40x does not have UART8
UART2_RX N18 UART2_RX
UART2_TX N17 UART2_TX
OUT_IO1 E8
OUT_IO2 E12
OUT_IO3 E6
OUT_IO4 E21
OUT_IO5 E7
OUT_IO6 E9
OUT_IO7 E23
OUT_IO8 E22
OUT_IO9 E11
OUT_IO10 E10
OUT_IO11 E24
OUT_IO12 E13
OUT_IO13 E20
OUT_IO1
OUT_IO2
OUT_IO3
OUT_IO4
OUT_IO5
OUT_IO6
OUT_IO7
OUT_IO8
OUT_IO9
OUT_IO10
OUT_IO11
OUT_IO12
OUT_IO13
USBP N3 USB_P
USBM N2 USB_N
USBID N1 USBID
VBUS N4 VBUS
SWCLK N8 SWCLK
SWDIO N7 SWDIO
SWO N6 SWO
nReset N9 nRST
BOOT0 N5 BOOT0
IO1 N16 IO1
IO2 N19 IO2
IO3 N21 IO3
IO4 N20 IO4
IO5 N23 IO5
IO6 E35 IO6
IO7 E36 IO7
IN_CRANK S22
IN_CAM S24
IN_VSS S25
IN_CRANK
IN_CAM
IN_VSS
IN_TPS S30
IN_TPS2 S32
IN_PPS S31
IN_PPS2 S33
IN_TPS
IN_TPS2
IN_PPS
IN_PPS2
IN_IAT S26
IN_AT1 S27
IN_CLT S28
IN_AT2 S29
IN_IAT
IN_AT1
IN_CLT
IN_AT2
V33_SWITCHABLE N28 V33_EXT
V33 N22 V33
V33 N32
E38
V5A_SWITCHABLE W2 V5A
V5A_SWITCHABLE
V5A_SWITCHABLE N30
GNDA W1
GNDA E39
IN_O2S S16
IN_O2S2 S15
IN_O2S
IN_O2S2
IN_RES1 S17
IN_RES2 S14
IN_RES3 S18
IN_RES1
IN_RES2
IN_RES3
IN_MAP1 S21
IN_MAP2 S20
IN_MAP3 S19
IN_MAP1
IN_MAP2
IN_MAP3
IN_AUX1 S13
IN_AUX2 S12
IN_AUX3 S11
IN_AUX4 S10
IN_AUX1
IN_AUX2
IN_AUX3
IN_AUX4
IN_SENS1 S9
IN_SENS2 S8
IN_SENS3 S7
IN_SENS4 S6
IN_KNOCK S23
IN_SENS1
IN_SENS2
IN_SENS3
IN_SENS4
IN_KNOCK
IN_D1 S4
IN_D2 S3
IN_D3 S2
IN_D4 S1
IN_D1
IN_D2
IN_D3
IN_D4
VREF1 S35
VREF2 S5
V33REF W3
VREF1
VREF2
V33_REF
IO8 N29
OUT_PWR_EN
VIGN N26
VBAT N27
IN_VIGN
VBAT
VCC N31
G
VCC
GND
Module:mega_mcu144/0.2
U_ign8
ign8.SchDoc
OUT_IGN1
OUT_IGN2
OUT_IGN3
OUT_IGN4
OUT_IGN5
OUT_IGN6
OUT_IGN7
OUT_IGN8
IGN1
IGN2
IGN3
IGN4
IGN5
IGN6
IGN7
IGN8
U_mcu
mcu.SchDoc
U_can
can.SchDoc
CAN_RX
CAN_TX
CANL
CANH
U_sd
sd.SchDoc
SD_SCK
SD_MISO
SD_MOSI
SD_CS
V33
V33_EXT
V33_PULL
U_sensors
sensors.SchDoc
ACC_SCK
ACC_MISO
ACC_MOSI
ACC_CS
BARO_SCL
BARO_SDA
SPI1_SCK
SPI1_MISO
SPI1_MOSI
SPI1_CS1
SPI1_CS2
nReset
nRST
S100
C147
100n
S101
BOOT0
I2C_SCL
I2C_SDA
SPI2_SCK
SPI2_MISO
SPI2_MOSI
SPI2_CS
SPI3_SCK
SPI3_MISO
SPI3_MOSI
SPI3_CS
USBP
USBM
USBID
VBUS
U_power
power.SchDoc
VBUS
VCC
PWR_EN
V5A
V33
V33A
V33_EXT
V33REF
IO8
R141
10k
P104
P105
P106
U_input
input.SchDoc
IN_CRANK
IN_CAM
IN_VSS
OUT_CRANK
OUT_CAM
OUT_VSS
Mux1
mux.SchDoc
IO0
IO1
COM
MUX
IN_TPS
IN_TPS2
OUT_TPS
OUT_TPS2
R142
10k
Mux2
mux.SchDoc
IN_PPS
IN_PPS2
OUT_PPS
OUT_PPS2
Mux3
mux.SchDoc
IN_IAT
IN_AT1
OUT_IAT
OUT_AT1
Mux4
mux.SchDoc
IN_CLT
IN_AT2
OUT_CLT
OUT_AT2
IN_O2S
IN_O2S2
OUT_O2S
OUT_O2S2
IN_RES1
IN_RES2
IN_RES3
OUT_RES1
OUT_RES2
OUT_RES3
IN_MAP1
IN_MAP2
IN_MAP3
OUT_MAP1
OUT_MAP2
OUT_MAP3
IN_AUX1
IN_AUX2
IN_AUX3
IN_AUX4
OUT_AUX1
OUT_AUX2
OUT_AUX3
OUT_AUX4
IN_SENS1
IN_SENS2
IN_SENS3
IN_SENS4
OUT_SENS1
OUT_SENS2
OUT_SENS3
OUT_SENS4
IN_KNOCK
OUT_KNOCK
VREF1
VREF2
U_dinput4
dinput4.SchDoc
IN_D1
IN_D2
IN_D3
IN_D4
OUT_D1
OUT_D2
OUT_D3
OUT_D4
V33
R144
4.7k
U110
CAN_RX
WAKEUP
SN74LVC1G06DCKR
VIGN
VBAT
IO1
IO2
IO3
IO4
IO5
IO6
IO7
IO8


More accurate threshold for "raw" connection of VR sensors
R195
5.1k
R196
4.7k
GNDA
OUT1
OUT2
U195
1
2
3
IN1 (CRNK)
V5A
4
Vdd
C195
100n
GNDA
IN3
5
6
7
Vss
14
13
12
IN2
11
GNDA
10
9
8
IN4
MCP6004
R197
OUT3
OUT4
GNDA
4.7k

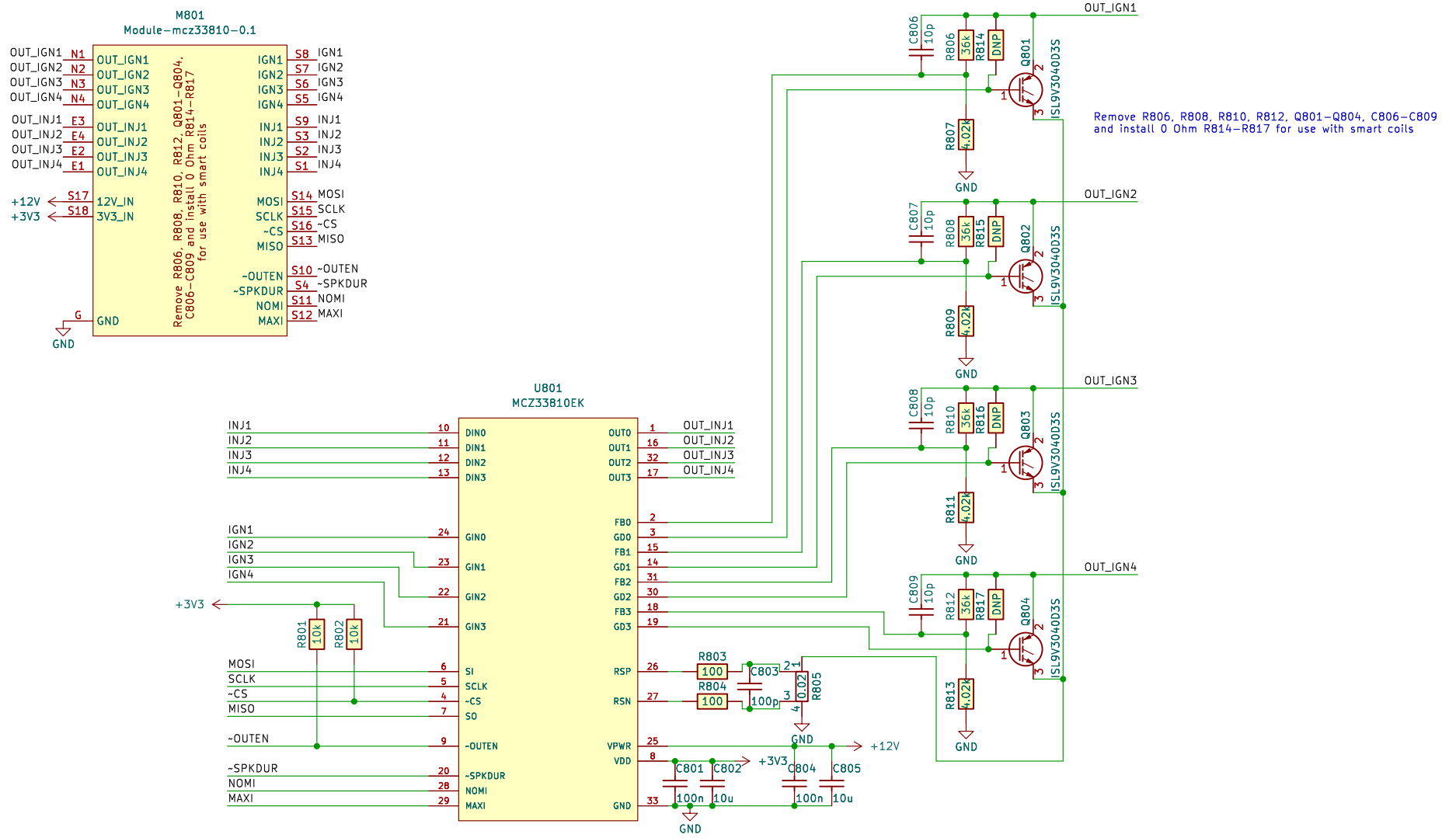


Title		
Size A4	Number	Revision
Date: 1.02.2024		Sheet of
File: C:\Work\..\power.SchDoc		Drawn By:


U106
LIS2DH12TR
ACC_SCK
ACC_CS
ACC_MISO
ACC_MOSI
SCL/SPC
CS
SDO/SA0
SDA/SDI/SDO
INT_1
INT_2
RES
GND(1)
VDD_IO
VDD
GND(3)
GND(2)
C126
100n
V33
C134
100n
U107
VDD
Vdd IO
INT DRDY
CS
SCL/SPC
SDA/SDI/SDO
SDO/SA0
RES
GND
GND
BARO_SCL
BARO_SDA
LPS22HBTR



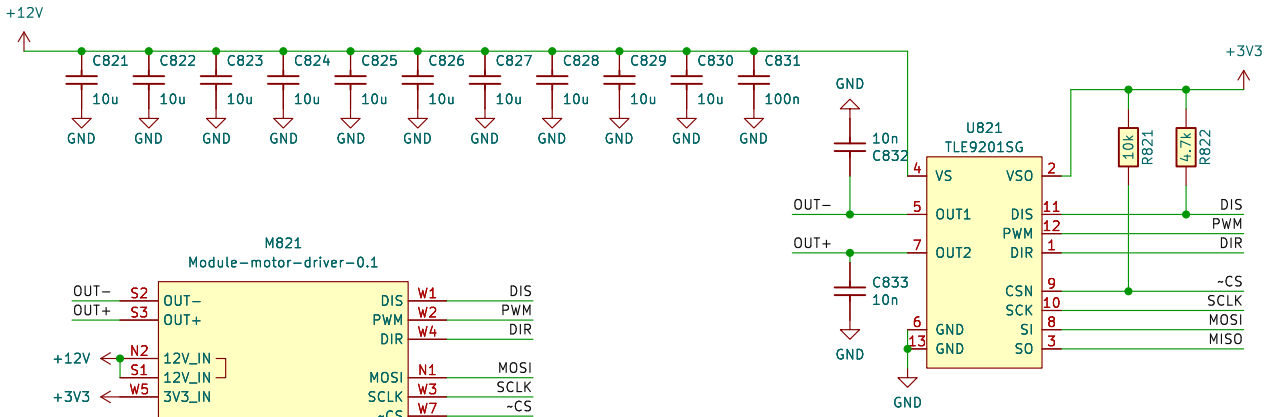
power_12and5V-0.2



Sheet: /
File: hellen1-mcz33810.kicad_sch

Title: MCZ33810 module

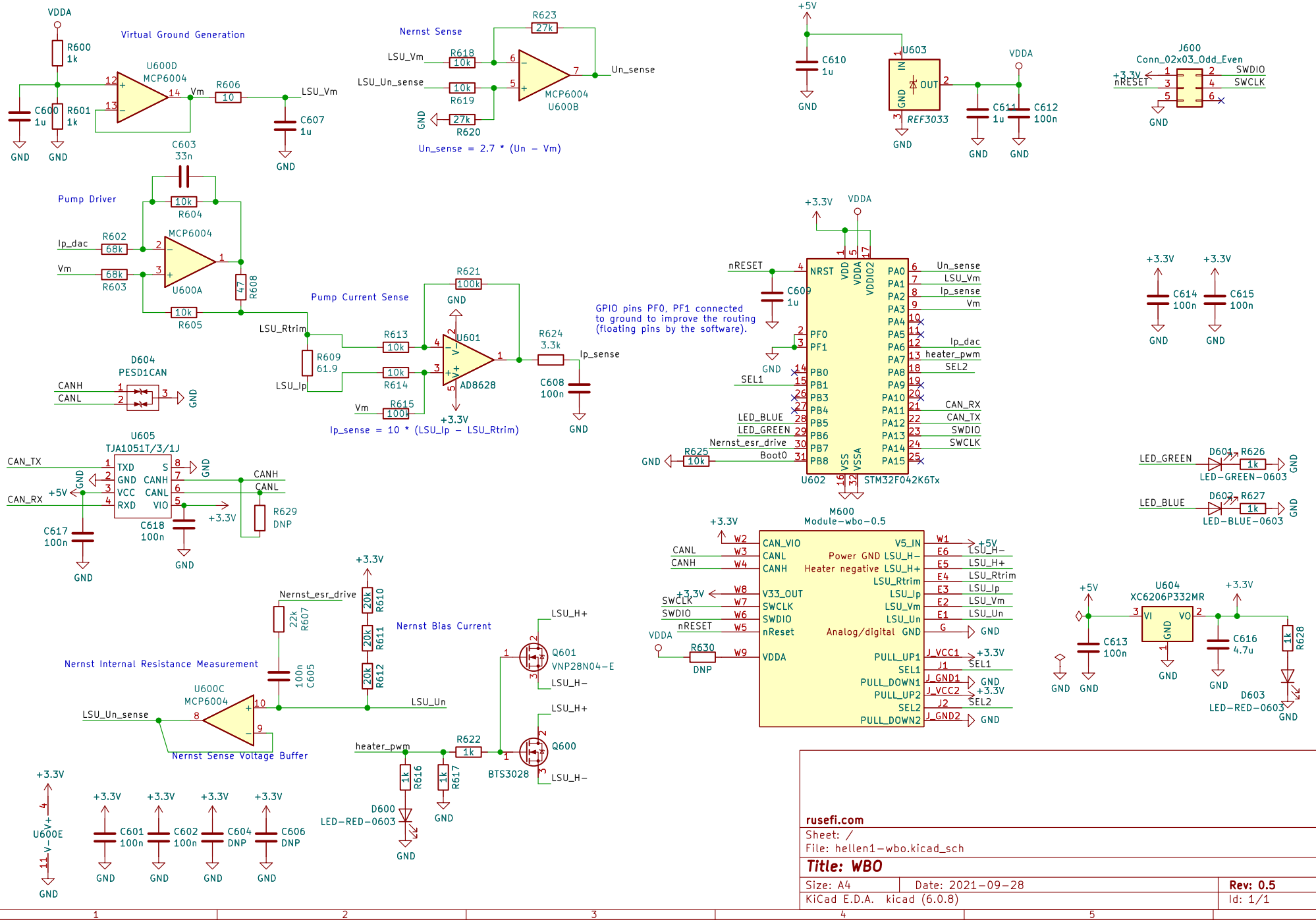
Size: A4	Date: 2024-03-07	**Rev: 0.1**
KiCad E.D.A. kicad 7.0.9		Id: 1/1

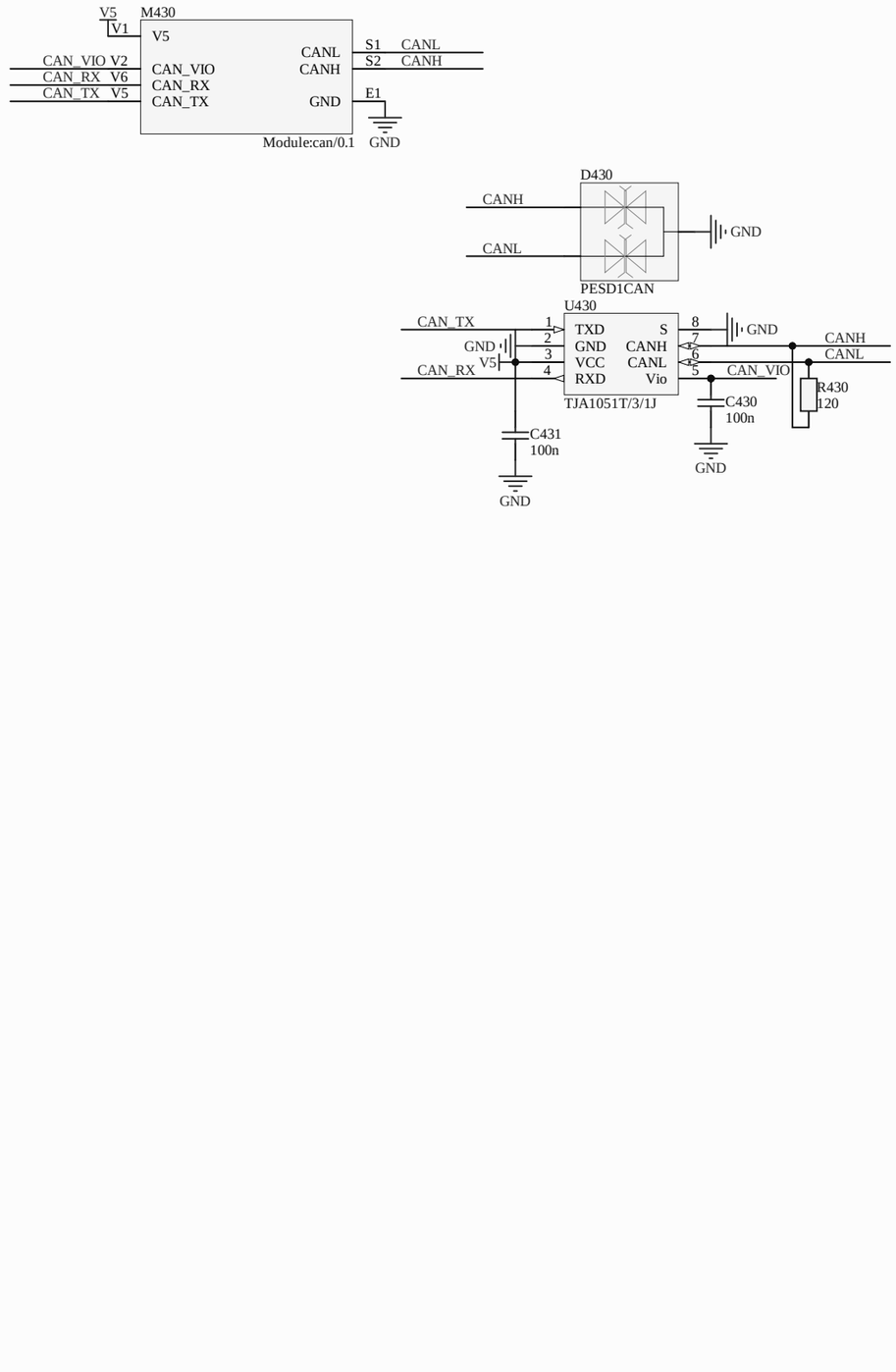


Sheet: /
File: hellen1−motor−driver.kicad_sch

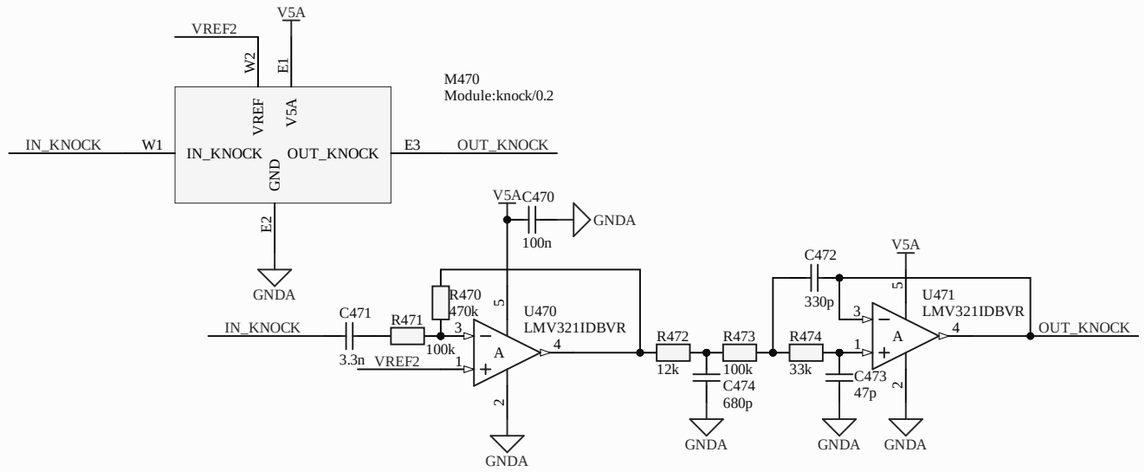
Title: Motor−driver module

Size: A4	Date: 2024−03−07	**Rev: 0.1**
KiCad E.D.A. 8.0.3		Id: 1/1


Virtual Ground Generation
U600D
MCP6004
R600 1k
R601 1k
C600 1u
C607 1u
R606 10
Vm
LSU_Vm
Nernst Sense
R623 27k
R618 10k
R619 10k
R620 27k
LSU_Vm
LSU_Un_sense
Un_sense
U600B
Un_sense = 2.7 * (Un − Vm)
+5V
C610 1u
U603
REF3033
C611 1u
C612 100n
VDDA
J600
Conn_02x03_Odd_Even
SWDIO
SWCLK
nRESET
Pump Driver
C603 33n
R604 10k
R605 10k
R602 68k
R603 68k
R608 47
Ip_dac
Vm
U600A
Pump Current Sense
R621 100k
R613 10k
R614 10k
R615 100
R609 61.9
R624 3.3k
C608 100n
U601
AD8628
LSU_Rtrim
LSU_Ip
Ip_sense
Ip_sense = 10 * (LSU_Ip − LSU_Rtrim)
D604
PESD1CAN
CANH
CANL
GPIO pins PF0, PF1 connected
to ground to improve the routing
(floating pins by the software).
nRESET
C609 1u
SEL1
LED_BLUE
LED_GREEN
Nernst_esr_drive
R625 10k
Boot0
U602
STM32F042K6Tx
C614 100n
C615 100n
U605
TJA1051T/3/1J
CAN_TX
CAN_RX
C617 100n
C618 100n
R629 DNP
LED_GREEN
LED_BLUE
D601
D602
R626 1k
R627 1k
LED−GREEN−0603
LED−BLUE−0603
M600
Module−wbo−0.5
Power GND LSU_H−
Heater negative LSU_H+
Analog/digital GND
R630 DNP
U604
XC6206P332MR
C613 100n
C616 4.7u
R628 1k
D603
LED−RED−0603
Nernst Internal Resistance Measurement
Nernst Bias Current
R607 22k
R610 20k
R611 20k
R612 20k
C605 100n
U600C
MCP6004
LSU_Un_sense
Nernst Sense Voltage Buffer
LSU_Un
Q601
VNP28N04−E
Q600
BTS3028
heater_pwm
R622 1k
R616 1k
R617 1k
D600
LED−RED−0603
U600E
C601 100n
C602 100n
C604 DNP
C606 DNP
rusefi.com
Sheet: /
File: hellen1−wbo.kicad_sch
Title: WBO
Size: A4
Date: 2021−09−28
Rev: 0.5
KiCad E.D.A. kicad (6.0.8)
Id: 1/1



Title: ***Hellen-One CAN***			***rusEFI EMS*** ***Hellen-One Platform*** *https://rusefi.com/s/hellen72*
Size: A3	Author: andreika	Revision: 0.1	
Date: 02.03.2021		Sheet 1 of 1	



Title: ***Hellen-One Knock***			***rusEFI EMS*** ***Hellen-One Platform*** *https://rusefi.com/s/hellen72*
Size: A3	Author: andreika	Revision: 0.2	
Date: 15.02.2023		Sheet 1 of 1	